

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 0606

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO4: K- Nearest Neighbour Learning – Locally weighted Regression – Radial Bases Functions – Case Based Learning

Assessment Tool Weightage: 20%

QUESTIONS: 05

Total

Marks:100

Annexure A

Industry Problem Solving

Code of Conduct:

I S. Dharani Teja_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Q1. Retail Product Recommendation Using KNN

The retail company can use **K-Nearest Neighbour (KNN)** to recommend products based on customers' previous purchases. Each customer is represented using features such as products purchased, purchase frequency, spending amount, product categories, and ratings.

1. Similarity Measurement

Similarity can be measured using **Euclidean distance**:

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

A smaller distance means that two customers have more similar purchasing behaviour.

For example, if Customer A and Customer B frequently purchase similar products, their feature vectors will have a small distance.

2. Recommendation Generation

1. Collect the customer's purchase history.
2. Convert customer behaviour into numerical features.
3. Calculate the distance between the target customer and all other customers.
4. Select the **K nearest customers**.
5. Identify products purchased by these similar customers but not yet purchased by the target customer.
6. Recommend the most frequently purchased or highly rated products.

For example, if 5 similar customers purchased a particular laptop accessory and the target customer has not purchased it, that accessory can be recommended.

3. Adapting to New Customer Data

KNN is a **lazy learning algorithm**, so it does not require complete retraining whenever new customer data arrives. New purchases can simply be added to the dataset. When a recommendation is requested, the algorithm considers the updated customer information.

Advantages

- Simple and easy to implement.
- Recommendations are based on actual customer behaviour.
- Can adapt easily to new purchase data.
- No complex training phase is required.

Conclusion

KNN provides personalized recommendations by identifying customers with similar purchasing patterns and recommending products based on their behaviour. This directly addresses the retail company's requirement to use past purchases for recommendation generation. The question specifically asks for similarity measurement, recommendation generation, and adaptation to new data.

Q2. Machine Output Prediction Using Locally Weighted Regression

A manufacturing company can use **Locally Weighted Regression (LWR)** to predict machine output from sensor readings such as temperature, pressure, vibration, speed, and operating time.

Unlike ordinary linear regression, LWR gives greater importance to training data points that are **closer to the current input**.

1. Working of Locally Weighted Regression

Suppose the current sensor readings are:

- Temperature = 75°C
- Pressure = 50 units
- Vibration = 10 units

LWR calculates the distance between this observation and historical observations.

A Gaussian weighting function can be used:

$$w_i = e^{-\frac{(x-x_i)^2}{2\tau^2}}$$

where:

- = weight of the training point
- = current sensor reading
- = historical sensor reading
- = bandwidth parameter

Nearby observations receive higher weights, while distant observations receive lower weights.

2. Local Models Improve Accuracy

The manufacturing process may behave differently under different operating conditions.

For example:

- Low temperature → normal machine output
- Medium temperature → increased output
- Very high temperature → reduced output

A single global regression model may not represent all these conditions accurately.

LWR instead creates a **local model around the current sensor condition**, making the prediction more suitable for the machine's current operating state.

3. Handling Noisy Data

Sensor readings can contain noise due to:

- Sensor errors
- Electrical interference
- Mechanical vibrations
- Temporary environmental changes

LWR reduces the influence of unrelated distant observations by assigning them smaller weights. Data preprocessing such as filtering, normalization, and outlier detection can further improve reliability.

4. Handling Incomplete Data

Missing sensor values can be handled using:

- Mean/median imputation
- Interpolation for time-series readings
- Using available sensor features
- Removing records with excessive missing values

Conclusion

Locally Weighted Regression is suitable because machine behaviour can change according to operating conditions. By giving more importance to nearby historical observations, the model can provide accurate local predictions while reducing the effect of irrelevant observations and noise.

Q3. Healthcare Risk Classification Using KNN

A healthcare company can use **K-Nearest Neighbour (KNN)** to classify patients into risk categories such as:

- Low Risk
- Medium Risk
- High Risk

Historical patient records can be used as training data.

1. Features Used

Relevant features may include:

- Age
- Blood pressure
- Cholesterol

- Blood glucose
- BMI
- Heart rate
- Previous medical measurements

The data should be normalized because different features may have different scales.

2. KNN Classification

For a new patient:

1. Collect the patient's medical features.
2. Normalize the features.
3. Calculate the distance between the new patient and historical patients.
4. Select the **K nearest patients**.
5. Identify their risk categories.
6. Assign the new patient the category receiving the majority vote.

For example, if and the nearest patients have:

- 3 High Risk
- 1 Medium Risk
- 1 Low Risk

the new patient is classified as **High Risk**.

3. Handling Large Datasets

KNN can become computationally expensive for large healthcare datasets because distances must be calculated against many records.

This can be improved using:

- KD-Trees
- Ball Trees
- Approximate nearest-neighbour methods
- Dimensionality reduction
- Efficient database indexing

4. Feature Selection

Not every medical feature contributes equally to risk prediction. Irrelevant features can negatively affect distance calculations.

Feature selection methods can be used to identify important attributes. For example, if blood pressure and glucose strongly influence risk while an unrelated feature does not, the important features should receive greater consideration.

5. Improving Accuracy

Accuracy can be improved by:

- Choosing an appropriate value of K
- Normalizing features
- Removing noisy data
- Handling missing values
- Selecting relevant features
- Using cross-validation

- Avoiding irrelevant attributes

Conclusion

KNN can classify new patients by comparing them with similar historical patients. Proper feature selection, preprocessing, and efficient search techniques make the model more practical for real-world healthcare datasets. The assessment specifically requires discussion of large datasets, feature selection, and accuracy improvement.

Q4. E-Commerce Dynamic Pricing Using Case-Based Learning

An e-commerce platform can use **Case-Based Learning (CBL)** to identify similar products and determine suitable prices based on previous pricing cases.

Each previous product situation is stored as a **case** containing information such as:

- Product category
- Brand
- Quality
- Demand
- Customer ratings
- Sales volume
- Competitor price
- Previous selling price
- Discount

1. Similarity Measurement

Similarity can be calculated by comparing the features of the new product with stored cases.

For numerical attributes, measures such as **Euclidean distance** can be used.

For example:

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

For categorical attributes such as product category or brand, matching-based similarity can be used.

A weighted similarity score can combine different features:

$$Similarity = \sum_{i=1}^n w_i s_i$$

where w_i represents the importance of each feature.

2. Reusing Past Cases

Case-Based Learning follows four major steps:

Retrieve → Reuse → Revise → Retain

1. **Retrieve:** Find products with similar characteristics.
2. **Reuse:** Use their historical pricing information.
3. **Revise:** Adjust the recommended price according to current demand, competition, and market conditions.
4. **Retain:** Store the new pricing case for future use.

For example, if a new smartphone is similar to previously sold smartphones, the system can retrieve those cases and use their historical prices as a starting point.

3. Scaling with Increasing Product Data

As the number of products increases, searching every stored case becomes expensive.

The system can scale using:

- Feature indexing
- Database indexing
- Clustering similar products
- Efficient nearest-neighbour search
- Removing outdated or duplicate cases
- Partitioning cases by category

Conclusion

Case-Based Learning is suitable because the e-commerce platform can directly reuse previous product and pricing experiences. As new products and transactions are processed, new cases can be retained, allowing the system to continuously improve its pricing decisions. This addresses the required similarity metrics, case reuse, and scalability aspects.

Q5. Delivery Time Forecasting Using RBF Network

A logistics company can use a **Radial Basis Function (RBF) Network** to predict delivery time based on factors such as:

- Delivery location

- Distance
- Traffic conditions
- Time of day
- Weather
- Vehicle type
- Historical delivery time

RBF networks are suitable because delivery time usually has a **non-linear relationship** with these factors.

1. RBF Network Structure

An RBF network generally contains:

Input Layer → RBF Hidden Layer → Output Layer

The input layer receives delivery-related features.

The hidden layer contains radial basis functions, commonly Gaussian functions:

$$\phi(x) = e^{-\frac{\|x-c\|^2}{2\sigma^2}}$$

where:

- X = input vector
- C = centre of the RBF
- σ = spread of the function

The output layer combines the RBF responses to produce the predicted delivery time.

2. Capturing Non-Linear Relationships

Traffic does not affect delivery time in a simple linear manner.

For example:

- Low traffic → small delay
- Moderate traffic → moderate delay
- Heavy traffic → very large delay

The RBF hidden layer can learn these local patterns and model complex relationships between location, traffic, and delivery time.

3. Improving Prediction Reliability

The system can continuously collect actual delivery times and compare them with predicted times.

Prediction reliability can be improved by:

- Updating the model using recent data
- Including real-time traffic information
- Removing noisy records
- Handling missing values
- Normalizing input features
- Monitoring prediction error

For changing traffic conditions, recent observations can be given greater importance during model updates.

Conclusion

An RBF Network is suitable for delivery-time forecasting because it can model complex non-linear relationships between traffic, location, and other factors. Continuous updates using new delivery data can improve prediction

reliability under changing conditions. This directly addresses the requirement to explain non-linear relationships and reliability under varying conditions.

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	25	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct Answer with no errors	Mostly correct with minor errors	Partially correct With noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand